Post-perturbation held-out DCLM loss, by model size

chinchilla 0.25 — adamw

chinchilla 0.25 — muon

chinchilla 0.5 — adamw

chinchilla 0.5 — muon

chinchilla 1 — adamw

chinchilla 1 — muon

chinchilla 2 — adamw

chinchilla 2 — muon

chinchilla 4 — adamw

chinchilla 4 — muon

chinchilla 8 — adamw

chinchilla 8 — muon

chinchilla 16 — adamw

chinchilla 16 — muon

chinchilla 32 — adamw

chinchilla 32 — muon

DCLM_heldout loss

model size (parameters)

30M 60M 100M 300M 600M

γ=0.003 γ=0.005 γ=0.007 γ=0.01 γ=0.02 γ=0.03 γ=0.04 γ=0.05 γ=0.07 γ=0.1 γ=0.13 unperturbed